F78PB Probability & Statistics B
Tutorial 1: Moment Generating Functions: Solutions

1. X has a density given by
$$f(x) = 2e^{-2x+2}, x \geq 1$$
and $f(x) = 0$ otherwise. One approach is to observe if $Y = X - 1$ then $Y \sim \text{Exp}(2)$ and then use that
$$M_Y(\theta) = \frac{2}{2-\theta}.$$
from the lecture notes or the "Distributions" formula sheet for $\theta < 2$. Now we need to recover the moment generating function for X:
$$M_X(\theta) = \mathbb{E}[e^{\theta X}] = \mathbb{E}[e^{\theta Y + \theta}] = e^{\theta} M_Y(\theta) = \frac{2e^{\theta}}{2-\theta}$$
Alternatively we use the density, f, of X directly to write So
$$\begin{aligned}
M_X(\theta) = \mathbb{E}[e^{\theta X}] &= \int_1^{\infty} e^{\theta x} f_X(x)\, dx, \\
&= \int_1^{\infty} e^{\theta x} \cdot 2e^{-2x+2}\, dx \\
&= 2e^2 \int_1^{\infty} e^{-(2-\theta)x} dx && \text{This integral is finite iff } \theta < 2. \\
&= 2e^2 \left[\frac{-e^{-(2-\theta)x}}{2-\theta} \right]_{x=1}^{x=\infty} \\
&= \frac{2e^2 e^{-2+\theta}}{2-\theta} = \frac{2e^{\theta}}{2-\theta}.
\end{aligned}$$
To show find $\mathbb{E}[X^2]$ from M_X we use the fact that
$$M_X''(0) = \mathbb{E}[X].$$
So differentiating M_X twice gives:
$$\begin{aligned}
M_X(\theta) &= 2e^{\theta}(2-\theta)^{-1}, \\
M_X'(\theta) &= 2e^{\theta}(2-\theta)^{-1} + 2e^{\theta}(2-\theta)^{-2} && \text{product rule} \\
M_X''(\theta) &= 2e^{\theta}(2-\theta)^{-1} + 4e^{\theta}(2-\theta)^{-2} + 4e^{\theta}(2-\theta)^{-3}.
\end{aligned}$$
Now setting $\theta = 0$ we have
$$M_X''(0) = 2(2)^{-1} + 4(2)^{-2} + 4(2)^{-3} = \frac{5}{2}.$$

2. (a) We have that X has density
$$f(x) = \frac{1}{2} e^{-|x|}.$$

Note this distribution is called the Laplace distribution. Now

$$\begin{aligned} M_X(\theta) &= \mathbb{E}[e^{\theta X}] \\ &= \int_{-\infty}^{\infty} e^{\theta x} f(x)\, dx \\ &= \frac{1}{2}\int_0^{\infty} e^{\theta x - x}\, dx + \frac{1}{2}\int_{-\infty}^{0} e^{\theta x + x} dx \\ &= \frac{1}{2}\int_0^{\infty} e^{-(1-\theta)x} + e^{-(1+\theta)x}\, dx. \end{aligned}$$

Note that this integral is finite if only if $\theta < 1$ and $\theta > -1$. Assuming $-1 < \theta < 1$ we have

$$\begin{aligned} \int_0^{\infty} e^{-(1-\theta)x} dx &= (1-\theta)^{-1}, \\ \int_0^{\infty} e^{-(1+\theta)x}\, dx &= (1+\theta)^{-1}, \end{aligned}$$

This follows from $x \mapsto \lambda e^{-\lambda x}$ being the pdf of an Exponential distribution so using $\lambda = 1-\theta$ and $\lambda = 1+\theta$ we get the above expressions. Therefore

$$M_X(\theta) = \frac{1}{2}\left((1-\theta)^{-1} + (1+\theta)^{-1}\right) = \frac{1+\theta+(1-\theta)}{2(1-\theta)(1+\theta)} = \frac{1}{(1-\theta)(1+\theta)} = \frac{1}{1-\theta^2}.$$

(b) To use the series expansion to find $\mathbb{E}[X^4]$ we need to find the expansion of $M_X(\theta)$. In this case $M_X(\theta)$ is the total of geometric series:

$$\sum_{k=0}^{\infty} a^k = \frac{1}{1-a}$$

with $a = \theta^2$. So

$$M_X(\theta) = \frac{1}{1-\theta^2} = 1 + \theta^2 + \theta^4 + \theta^6 + \dots$$

In general we have

$$M_X(\theta) = 1 + \mathbb{E}[X]\theta + \frac{\mathbb{E}[X^2]}{2!}\theta^2 + \frac{\mathbb{E}[X^3]}{3!}\theta^3 + \frac{\mathbb{E}[X^4]}{4!}\theta^4 + \dots$$

So equating powers of θ we have

$$\mathbb{E}[X^4] = 4! = 24.$$

3. Let $X \sim \text{Gamma}(t, \lambda)$ with $t, \lambda > 0$.

(a) By definition,

$$\begin{aligned} M_X(\theta) = \mathbb{E}[e^{\theta X}] &= \int_0^\infty e^{\theta x} f_X(x) dx \\ &= \int_0^\infty e^{\theta x} \frac{\lambda^t x^{t-1} e^{-\lambda x}}{\Gamma(t)}\, dx \\ &= \frac{\lambda^t}{\Gamma(t)} \int_0^\infty x^{t-1} e^{-(\lambda-\theta)x} dx. \end{aligned}$$

Note this integral is finite if and only if $\theta < \lambda$. Now we can rewrite this integral in terms of a Gamma density

$$M_X(\theta) = \frac{\lambda^t}{(\lambda-\theta)^t} \int_0^\infty \frac{(\lambda-\theta)^t}{\Gamma(t)} x^{t-1} e^{-(\lambda-\theta)x} dx = \frac{\lambda^t}{(\lambda-\theta)^t}.$$

(b) For $X \sim \text{Gamma}(2, t)$ we use part (a) to write

$$M_X(\theta) = \frac{1}{\left(1 - \frac{\theta}{\lambda}\right)^2}.$$

We want to expand this up to order θ^4 terms, first we use a geometric series

$$\frac{1}{\left(1 - \frac{\theta}{\lambda}\right)} = 1 + \frac{\theta}{\lambda} + \frac{\theta^2}{\lambda^2} + \frac{\theta^3}{\lambda^3} + \frac{\theta^4}{\lambda^4} + \dots$$

If we square this and expand we obtain the series for $M_X(\theta)$

$$\begin{aligned} M_X(\theta) &= \left(1 + \frac{\theta}{\lambda} + \frac{\theta^2}{\lambda^2} + \frac{\theta^3}{\lambda^3} + \frac{\theta^4}{\lambda^4} + \dots\right)\left(1 + \frac{\theta}{\lambda} + \frac{\theta^2}{\lambda^2} + \frac{\theta^3}{\lambda^3} + \frac{\theta^4}{\lambda^4} + \dots\right) \\ &= 1 + \frac{2\theta}{\lambda} + \frac{3\theta^2}{\lambda^2} + \frac{4\theta^3}{\lambda^3} + \frac{5\theta^4}{\lambda^4} + \dots \end{aligned}$$

We compare this to the expansion

$$M_X(\theta) = 1 + \mathbb{E}[X]\theta + \frac{\mathbb{E}[X^2]}{2!}\theta^2 + \frac{\mathbb{E}[X^3]}{3!}\theta^3 + \frac{\mathbb{E}[X^4]}{4!}\theta^4 + \dots$$

Equating the θ^4 terms we have

$$\frac{\mathbb{E}[X^4]}{4!} = \frac{5\theta^4}{\lambda^4}.$$

Rearranging give $\mathbb{E}[X^4] = \frac{5!}{\lambda^4}$.

(c) Suppose $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Gamma}(t, \lambda)$ and $Z = \sum X_i$. Then

$$\begin{aligned} M_Z(\theta) &= \mathbb{E}[e^{\theta Z}] = \mathbb{E}[e^{\theta X_1 + \ldots + \theta X_n}] \\ &= \mathbb{E}\left[\prod_{i=1}^{n} e^{\theta X_i}\right] \\ &= \prod_{i=1}^{n} \mathbb{E}[e^{\theta X_i}] && \text{by independence} \\ &= \prod_{i=1}^{n} M_{X_i}(\theta) \\ &= M_X(\theta)^n \\ &= \frac{\lambda^{nt}}{(\lambda - \theta)^{tn}} \end{aligned}$$

By comparing with the answer to part (a) we see this is the moment generating function of a Gamma(nt, λ) distributed random variable, so by the uniqueness property of moment generating functions we have $Z \sim \text{Gamma}(nt, \lambda)$.

4. Suppose $X \sim \text{Gamma}(t, \lambda)$ and $Y = cX$ then

$$M_X(\theta) = \mathbb{E}[e^{\theta Y}] = \mathbb{E}[e^{c\theta X}] = M_X(c\theta).$$

By Q3(a) or from the "Distributions" formula sheet we have

$$M_X(\theta) = \frac{\lambda^t}{(\lambda - \theta)^t}$$

for $\theta < t$. Therefore

$$M_Y(\theta) = M_X(c\theta) = \frac{\lambda^t}{(\lambda - c\theta)^t} = \frac{(\lambda/c)^t}{(\lambda/c - \theta)^t}.$$

Observe that this is the moment generating function of a Gamma distribution, in particular a Gamma$(t, \lambda/c)$ distribution by the uniqueness property of moment generating functions $Y \sim \text{Gamma}(t, \lambda/c)$.

5. Suppose that $X \sim \text{N}(3, 1)$ then

$$M_X(\theta) = e^{3\theta} e^{\theta^2/2}.$$

Let us find the expansion:

$$\begin{aligned} e^{3\theta} &= 1 + 3\theta + \frac{3^2\theta^2}{2!} + \frac{3^3\theta^3}{3!} + \frac{3^4\theta^4}{4!} + \ldots \\ e^{\theta^2/2} &= 1 + \frac{\theta^2}{2} + \frac{\theta^4}{(2!)(2^2)} + \ldots \end{aligned}$$

Multiplying these together

$$\begin{aligned} M_X(\theta) &= \left(1 + 3\theta + \frac{3^2\theta^2}{2!} + \frac{3^3\theta^3}{3!} + \frac{3^4\theta^4}{4!} + \dots\right)\left(1 + \frac{\theta^2}{2} + \frac{\theta^4}{(2!)(2^2)} + \dots\right) \\ &= 1 + 3\theta + (\frac{3^2}{2} + \frac{1}{2})\theta^2 + \left(\frac{3^3}{3!} + \frac{3}{2}\right)\theta^3 + \left(\frac{1}{2!(2^2)} + \frac{3^2}{(2)(2!)} + \frac{3^4}{4!}\right)\theta^4 + \dots \\ &= 1 + 3\theta + 5\theta^2 + 6\theta^3 + \left(\frac{23}{4}\right)\theta^4 + \dots \end{aligned}$$

We compare this to the expansion

$$M_X(\theta) = 1 + \mathbb{E}[X]\theta + \frac{\mathbb{E}[X^2]}{2!}\theta^2 + \frac{\mathbb{E}[X^3]}{3!}\theta^3 + \frac{\mathbb{E}[X^4]}{4!}\theta^4 + \dots$$

Equating terms we have

$$\begin{aligned} \theta^3 : &\qquad \frac{\mathbb{E}[X^3]}{3!} = 6 \\ \theta^4 : &\qquad \frac{\mathbb{E}[X^4]}{4!} = \frac{23}{4}. \end{aligned}$$

Rearranging gives $\mathbb{E}[X^3] = 36, \mathbb{E}[X^4] = 138$.

6. Suppose that $X \sim \mathrm{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathrm{N}(\mu_Y, \sigma_Y^2)$ are independent then from the lecture notes or the "Distributions" formula sheet

$$\begin{aligned} M_X(\theta) &= e^{\mu_X\theta + \frac{1}{2}\sigma_X^2\theta^2} \\ M_Y(\theta) &= e^{\mu_Y\theta + \frac{1}{2}\sigma_Y^2\theta^2}. \end{aligned}$$

Now let $S = aX + bY$ then by independence

$$\begin{aligned} M_S(\theta) &= \mathbb{E}[e^{a\theta X + b\theta Y}] \\ &= \mathbb{E}[e^{a\theta X}]\mathbb{E}[e^{b\theta Y}] \\ &= M_X(a\theta)M_Y(b\theta) \\ &= e^{a\mu_X\theta + \frac{1}{2}a^2\sigma_X^2\theta^2} e^{b\mu_Y\theta + \frac{1}{2}b^2\sigma_Y^2\theta^2} \\ &= e^{(a\mu_X + b\mu_Y)\theta + \frac{1}{2}(a^2\sigma_X^2 + b^2\sigma_Y^2)\theta^2}. \end{aligned}$$

This is the moment generating function for a $\mathrm{N}(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2)$ so by the uniqueness property of moment generating functions $S \sim \mathrm{N}(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2)$.

Application Let $S = Y - X$ time between Anne arriving and David arriving, for $X \sim \mathrm{N}(0, 3^2)$ the time Anne arrives and $Y \sim \mathrm{N}(1, 4^2)$ the time David arrives. Then the event that Anne waits more than 4 minutes for David is the event $\{S > 4\}$.

By the earlier part of the question we have $S \sim \mathrm{N}(1, 3^2 + 4^2)$ (note $3^2 + 4^4 = 5^2$ so S has standard deviation 5) then

$$\begin{aligned}
\mathbb{P}(S > 4) &= \mathbb{P}(\frac{S-1}{5} > \frac{4-1}{5}) = \mathbb{P}(Z > 0.6) \\
&= 1 - \mathbb{P}(Z \leq 0.6) && \text{for } Z \sim \mathrm{N}(0,1) \\
&= 1 - \Phi(0.6) \\
&= 1 - 0.7257 \\
&= 0.2743.
\end{aligned}$$

7. Let $X_1, X_2, \ldots$ be i.i.d. with common moment generating function M_X, and suppose that N is an independent counting variable with probability generating function G_N. Let $S_N = X_1 + \ldots + X_N$. Using the partition theorem for expectation we can write the moment generating function of S_N as

$$\begin{aligned}
M_{S_N}(\theta) &= \mathbb{E}[e^{\theta S_N}] \\
&= \sum_{n=0}^{\infty} \mathbb{E}[e^{\theta S_N} | N = n] \mathbb{P}(N = n) \\
&= \sum_{n=0}^{\infty} M_{S_n}(\theta) \mathbb{P}(N = n)
\end{aligned}$$

Now we need to find the moment generating function for S_n:

$$M_{S_n}(\theta) = \mathbb{E}[e^{\sum_i \theta X_i}] = \mathbb{E}[\prod_i e^{\theta X_i}].$$

By independence

$$M_{S_n}(\theta) = \prod_i \mathbb{E}[e^{\theta X_i}] = M_X(\theta)^n.$$

Now using this expression in our expression for M_{S_N} gives

$$M_{S_N}(\theta) = \sum_{n=0}^{\infty} M_X(\theta)^n \mathbb{P}(N = n)$$

Let $s = M_X(\theta)$ (noting this is a deterministic quantity that does not depend on n) then

$$M_{S_N}(\theta) = \sum_{n=0}^{\infty} s^n \mathbb{P}(N = n).$$

Using that for a discrete random variable N and function g we can write

$$\mathbb{E}[g(N)] = \sum_{n \in \mathrm{Im}(N)} g(n) \mathbb{P}(N = n)$$

we observe that

$$G_N(s) = \mathbb{E}[s^N] = \sum_{n=0}^{\infty} s^n \mathbb{P}(N = n).$$

Therefore

$$M_{S_N}(\theta) = G_N(s) = G_N(M_X(\theta)).$$

Alternatively we could recall this fact from the lecture notes.

Now applying the chain rule

$$M'_{S_N}(\theta) = G'_N(M_X(\theta))M'_X(\theta).$$

Evaluating this at $\theta = 0$ gives

$$M'_{S_N}(0) = G'_N(M_X(0))M'_X(0).$$

Since $M_X(0) = 1$ this is

$$M'_{S_N}(0) = G'_N(1)M'_X(0).$$

By the properties of moment generating and probability generating functions

$$\mathbb{E}[S_N] = M'_{S_N}(0) = G'_N(1)M'_X(0) = \mathbb{E}[N]\mathbb{E}[X].$$

8. Let $N \sim \text{Geo}\,(1/5)$ be the number of calls needed to receive a correct answer. Let $X_i \sim \text{Exp}(2)$ be the time between the $(i-1)^{th}$ call and the ith call. So $X_1, X_2 \ldots$ are i.i.d with common moment generating function M_X. Then let $S = \sum_{i=1}^{N} X_i$ be the time until the first successful call.

   From the notes

   $$M_{S_N}(\theta) = G_N(M_X(\theta)).$$

   From the lecture notes or the "Distributions" formula sheet

   $$M_X(\theta) = \frac{2}{2-\theta} \qquad \text{for } \lambda < 2.$$

   We can calculate the probability generating function for N as follows: Let $p = 1/5$, then

   $$\begin{aligned}
   \mathbb{E}[N] &= \sum_{n=1}^{\infty} s^n \mathbb{P}(N = n) \\
   &= \sum_{n=1}^{\infty} s^n p(1-p)^{n-1} \\
   &= ps \sum_{n=1}^{\infty} (s - sp)^{n-1}
   \end{aligned}$$

We recognise this as a geometric series which converges for any $|s| < (1-p)^{-1}$ to give

$$\mathbb{E}[N] = \frac{ps}{1-s(1-p)} = \frac{s}{5-4s}.$$

Now

$$M_{S_N}(\theta) = G_N(M_X(\theta)) = \frac{\frac{2}{2-\theta}}{5-4\left(\frac{2}{2-\theta}\right)} = \frac{2}{10-5\theta-8} = \frac{2}{2-5\theta} = \frac{2/5}{2/5-\theta}.$$

We observe this is the moment generating function for the Exp(2/5)-distribution and hence $S_N \sim \text{Exp}(2/5)$ by the uniqueness property of the moment generating function.

9. (a) Y is a discrete random variable and admits a valid probability mass function (pmf) if X does. To see that X admits a valid pmf we need to check that the probabilities $w, 1-2w, w$ are non-negative and sum to 1. We observe that

$$w + 1 - 2w + w = 1$$

for all choices of w so it remains to check that all the probabilities are non-negative. That is we require

$$w \geq 0 \quad \text{and } 1 - 2w \geq 0.$$

Rearranging this gives

$$0 \leq w \leq \frac{1}{2}.$$

So X (and hence Y) admits a valid density for any value $w \in [0, 1/2]$.

(b) Before calculating the cumulant generating function we first determine the moment generating function

$$\begin{aligned} M_Y(\theta) &= M_X(a\theta) = \mathbb{E}[e^{a\theta X}] \\ &= e^{-a\theta}\mathbb{P}(X=-1) + \mathbb{P}(X=0) + e^{a\theta}\mathbb{P}(X=1) \\ &= we^{-a\theta} + 1 - 2w + we^{a\theta}. \end{aligned}$$

Therefore the cumulant generating function is

$$K_Y(\theta) = \log(M_Y(\theta)) = \log\left(1 - 2w + w(e^{a\theta} + e^{-a\theta})\right).$$

(c) Note that differentiating K_Y n times gives $\kappa_n = K_Y^{(n)}(0)$ where is the n-cumulant. Since $\kappa_1 = \mathbb{E}(Y)$ and $\kappa_2 = \mathbb{V}\text{ar}(Y)$ we will differentiate the cumulant generating function twice:

$$\begin{aligned} K_Y'(\theta) &= \frac{aw(e^{a\theta} - e^{-a\theta})}{1 - 2w + w(e^{a\theta} + e^{-a\theta})} \\ K_Y''(\theta) &= \frac{a^2 w(e^{a\theta} + e^{-a\theta})}{1 - 2w + w(e^{a\theta} + e^{-a\theta})} - \frac{(aw(e^{a\theta} - e^{-a\theta}))^2}{(1 - 2w + w(e^{a\theta} + e^{-a\theta}))^2} \end{aligned}$$

Setting $\theta = 0$

$$\begin{aligned} \mathbb{E}(Y) = K_Y'(0) &= \frac{aw(1-1)}{1 - 2w + w(1+1)} = 0 \\ \mathbb{V}\text{ar}(Y) = K_Y''(0) &= \frac{2a^2 w}{1 - 2w + 2w} = 2a^2 w. \end{aligned}$$

(d) Since Y is symmetric the third cumulant is zero and hence $\gamma_1 = 0$.

10. Let X_1, X_2 be two independent random variables. Then the moment generating function of the sum $Y = X_1 + X_2$ is given by

$$M_Y(\theta) = M_{X_1}(\theta) M_{X_2}(\theta)$$

by independence. Now taking the logarithm

$$\begin{aligned} K_Y(\theta) &= \log(M_Y(\theta)) = \log(M_{X_1}(\theta) M_{X_2}(\theta)) \\ &= \log(M_{X_1}(\theta)) + \log(M_{X_2}(\theta)) = K_{X_1}(\theta) + K_{X_2}(\theta). \end{aligned}$$

Exam Style Question

(a) We will use the property

$$\mathbb{E}[X^k] = M_X^{(k)}(0).$$

(1 mark)

Using the expression we have for $M_X(\theta)$

$$\begin{aligned} M_X(\theta) &= (1 - 10\theta)^{-2} \\ M_X'(\theta) &= 20(1 - 10\theta)^{-3} \\ M_X''(\theta) &= 600(1 - 10\theta)^{-4} \\ M_X'''(\theta) &= 24000(1 - 10\theta)^{-5}. \end{aligned}$$

(2 marks)

Setting $\theta = 0$ gives

$$\begin{aligned} \mathbb{E}[X^2] &= M_X''(0) = 600 \\ \mathbb{E}[X^3] &= M_X'''(0) = 24000. \end{aligned}$$

(1 mark)

(b) (i) Using the definition of the moment generating function and $Y = -\log(W)$ we have

$$M_Y(\theta) = \mathbb{E}[e^{\theta Y}] = \mathbb{E}[e^{-\theta \log(W)}] = \mathbb{E}[W^{-\theta}].$$

(2 marks)

Since $W \sim \mathrm{U}(0,1)$ it has probability density function $f_W(w) = 1$ for $w \in [0,1]$ and $f_W(w) = 0$ otherwise, therefore for $\theta < 1$

$$\mathbb{E}[W^{-\theta}] = \int_0^1 w^{-\theta} dw = \left[\frac{w^{1-\theta}}{1-\theta}\right]_{w=0}^1 = \frac{1}{1-\theta}.$$

Note that this integral is finite only for $\theta < 1$. (1 mark)

This is the moment generating function of Exp(1) so by uniqueness the distribution of Y is Exp(1). (1 mark)

(ii) Note that

$$\mathbb{E}[(\log(W))^2] = \mathbb{E}[Y^2] = M_Y''(0).$$

(1 mark)

Now

$$M_Y'(\theta) = (1-\theta)^{-2}$$
$$M_Y''(\theta) = 2(1-\theta)^{-3}.$$

Therefore $\mathbb{E}[Y^2] = 2$. (1 mark)